Simple and compound gear trains

ENGINEERING WITH MR KELLY

National 5 · Mechanisms & structures · Class worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ calculate velocity ratios for simple and compound trains
- ☐ calculate input and output speeds
- ☐ state directions of rotation
- ☐ give reasons for choosing plastic gears

I DO — Key ideas and worked example

Read this first

VR = driven ÷ driver (teeth)

Velocity ratio = teeth on driven gear ÷ teeth on driver gear = input speed ÷ output speed

- Output speed = input speed ÷ VR
- Speeds in revs min^{-1}

- In a **compound** gear train two gears are fixed on the same shaft and turn at the same speed.
- **Total VR = VR_1 × VR_2**, where VR_1 = B ÷ A and VR_2 = D ÷ C.
- Compound trains give a large speed change in a small space.
- Simple train: VR = driven ÷ driver. Meshing gears turn in opposite directions.

A winch uses a compound gear train: A = 12, B = 48, C = 18, D = 54 teeth. The drum on D turns at 50 revs min^{-1}.

Example Calculate the velocity ratio and the motor speed needed.

Worked answer set out every step like this

VR = (48 ÷ 12) × (54 ÷ 18) = 4 × 3 = 12

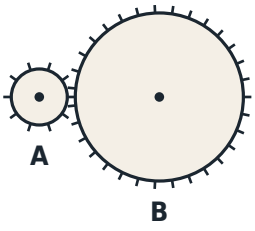
Input speed = output speed × VR

Motor speed = 50 × 12 = 600 revs min^{-1}

YOU DO — On your own

Show your working. Check your answers afterwards.

1 *A simple gear train: driver A = 15 teeth, driven B = 45 teeth. A rotates at 900 revs min^{-1}.*



a) Calculate the velocity ratio.

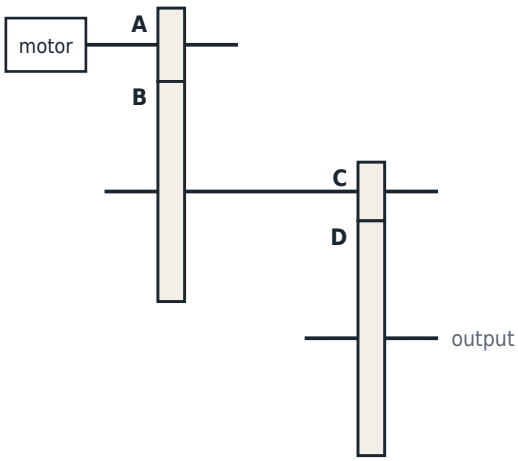
b) Calculate the speed of B.

2 *A conveyor uses a simple gear train: A = 20 teeth drives B = 60 teeth at 720 revs min^{-1}.*

a) Calculate the velocity ratio.

b) Calculate the output speed.

3 *A compound gear train: A = 25, B = 75, C = 20, D = 80 teeth. A turns at 1000 revs min^{-1}.*



a) Calculate the velocity ratio.

b) Calculate the output speed of D.

4 *A simple gear train: a 10-tooth driver turns a 40-tooth driven gear at 600 revs min^{-1}.*

a) Calculate the velocity ratio.

b) Calculate the output speed.

c) State the direction of the output compared with the driver.

5 *A small lathe uses a compound gear train: A = 16, B = 64, C = 24, D = 72 teeth. A turns at 1500 revs min^{-1}.*
a) Calculate the velocity ratio.

b) Calculate the output shaft speed.

6 *A simple gear train is designed to increase speed: A = 30 teeth, B = 10 teeth. A turns at 300 revs min^{-1}.*
a) Calculate the velocity ratio.

b) Calculate the speed of B.

7 *A compound gear train: A = 18, B = 72, C = 24, D = 60 teeth. The input turns at 1200 revs min^{-1}.*
a) Calculate the velocity ratio.

b) Calculate the output speed.

8 *A model vehicle uses a 12-tooth driver and a 36-tooth driven gear. The driver turns at 1800 revs min^{-1}.*
a) Calculate the velocity ratio.

b) Calculate the speed of the driven gear.

c) State one reason for using plastic gears in the model.

9 *A hoist uses a compound gear train: A = 14, B = 70, C = 20, D = 80 teeth. The drum on D turns at 20 revs min^{-1}.*

a) Calculate the overall velocity ratio.

b) Calculate the motor speed needed.

Check and reflect

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.